Telematics Platform User Guide

Update Package Generation ConMod

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1 Introduction

This document provides usage information for Update Package Generation. The feature is part of two other features:

- Telematics Platform Software Loading framework that includes processes running on the applications processor within the System on Chip of the Network Access Device of a Telematics Control Unit.
- The Software Development Kit for the Telematics Platform.

This guide focuses on providing an overview on how the Project should use the tools provided in the Software Development Kit to generate an update package file to be used for the Software Loading framework.

The definition of an update package is explained with detail in TP_F101U-00028.

The definition of content type is explained with detail in TP_F100W-00015.

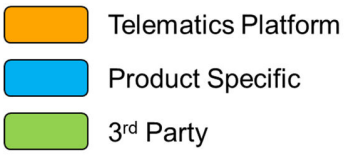
The installation of the Software Development Kit is explained with detail in TP_F101U-00035.

1.1 Notational Conventions

Ctrl+x	*A sequence with a '+' indicates that you must press and hold one key or pointing device button while pressing another key or pointing device button*
[ ]	*Optional elements in a command*
…..	*Indicates one of the following possibilities:* • *Additional optional arguments in a statement have been omitted* • *The preceding item or items can be repeated one or more times* • *Additional parameters, values, or other information can be entered*
-	*Line continuation: indicates that a command line or code line continues on the following line when at the end of a line*
Italic text	*Indicates important information, complete titles of manuals or variables*
`Mono-spaced`	*Example code or interactive screen display/output*
Numbers	*All numbers are assumed to be decimal unless otherwise noted. Non decimal radixes – binary, octal or hexadecimal – are explicitly indicated*
Underline blue text	*A link to an external reference (e.g., website, document)*

`Modules`
`|-> section`
`|-> item`
	A link to a section or item in the Telematics Platform html interface specification (Doxygen output from platform code headers)

Diagram Legend



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2 Setup

2.1 Tools/Equipment Needed

The required tools should be included as part of the Software Development Kit for the Telematics Platform. Follow the instructions provided by TP_F101U-00035 in order to set the appropriate environment. The tools provided for the Update Package Generation process can run only in machines with Linux-based Operation Systems. In particularly, it has only been executed in Ubuntu 16.04.6 LTS but it should not be limited to that Operating System.

2.2 Setup Instructions

The details about the setup of the different hardware that can use the Software loading Manager or required for the development of a particular Update Agent or Download Client is beyond the scope of this guide. The details about the installation and usage of the Software Development Kit and the compilation of binaries is beyond the scope of this guide as well. This document just covers instructions to successfully generate an Update Package in a Linux-based Operating System. For that, the following component(s) should be installed in the host computer:

- /bin/bash executable

2.3 Setup Troubleshooting

Not applicable

3 Features/Components

3.1 Software Download Blocks Generation

3.1.1 Overview

3.1.1.1 Platform Content Overview

Not Applicable

3.1.1.2 Product Expectations

Not Applicable

3.1.1.3 Extending Platform Functionality

Not Applicable

3.1.2 Setup

3.1.2.1 Tools/Equipment Needed

Not Applicable

3.1.2.2 Setup Instructions

Not Applicable

3.1.2.3 Setup Troubleshooting

Not Applicable

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3.1.3 Usage

3.1.3.1 Basic Instructions

Not Applicable

3.1.3.2 Detailed instructions

!Note: The purpose of this section is to replace the Section 3.2.3.2 in Update Package Generation User Guide (TP_F101U-00040).

3.1.3.2.1 Parameters

The script accepts five parameters, two of them are mandatory and the other ones are optional. If an optional parameter is not provided, the default value will be used.

These are the script parameters:

Parameter	*Description*	*Optional*	*Default Value*
Images Directory (-i)	Folder that contains the images	No	N/A
Output Directory (-o)	Folder in which the SWDL block will be placed.	No	N/A
Blocks Config File (-b)	JSON Configuration file that contains the information about the blocks content and mapping between image and component names.	Yes	<CAS_WORK_DIR>/.build/config/swdl_blocks.json
SWL Scripts Directory (-s)	Folder in which the Update Package Generation Scripts are located.	Yes	<CAS_WORK_DIR>/package/conti/conti-swl/scripts/
Environment Configuration File (-e)	Config file generated by the set_env.sh execution. See section 3.1.3.2.1.	Yes	<CAS_WORK_DIR>/up_environment

3.1.3.2.2 Blocks config file

The script requires an input configuration file in JSON format. This format consists in “block_name” and “content” attributes for each of the blocks as well as mapping between image and component names.

Attribute	*Description*
block_name	This will be used to name the SWDL block
content	A set of image filenames mapped to one or more component names that will be included in the SWDL block. The files must be in the images directory that is given as a parameter. Also if the component name is empty (e.g., {<image_name_4> ": “” }), then the component is not included in CRC and version checks.

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Here is an example:

```
[
  {
    "block_name": "Block_1",
    "content": [
      {"<image_name_1>": "<component_name_1_1>"},
      {"<image_name_2>": "<component_name_2_1>"},
      {"<image_name_3>": "<component_name_3_1>"},
      {"<image_name_4>": ""}, //component is not included in CRC and version check
      {"<image_name_5>": [
          "<component_name_5_1>"
          "<component_name_5_2>"
          "<component_name_5_3>"
        ]
      },
      {"<image_name_6>": "<component_name_6_1>"}
    ]
  },
  {
    "block_name": "Block_2",
    "content": [
      {"<image_name_1>": [
          "<component_name_1_1>",
          "<component_name_1_2>",
          "<component_name_1_3>",
          "<component_name_1_4>",
          "<component_name_1_5>",
          "<component_name_1_6>",
          "<component_name_1_7>"
        ]
      },
      {"<image_name_2>": "<component_name_2_1>"},
      {"<image_name_3>": "<component_name_3_1>"}
    ]
  },
  {
    "block_name": "Block_3",
    "content": [
      {"<image_name_1>": "<component_name_1_1>"},
      {"<image_name_2>": "<component_name_2_1>"},
    ]
  }
```

Figure 14 SWDL blocks JSON file

As reference, attached is a swdl blocks config file that can be used as a template:

swdl_blocks.json

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!Notes:

i. If the update campaign is intended for multiple processors, there must be one JSON config file for each of the processors.
ii. The file attached is only a template and must be modified according to the project needs.

3.1.3.2.3 Executing the script

This script allows the user to execute it in two ways:

- From the console itself
- As an imported python module

Here is an example of how to execute it from the console with no optional parameters:

```
python generate_swdl_blocks.py -o /Shared/output/ -i /Shared/4K/
```

And with all the parameters:

```
python generate_swdl_blocks.py -o /Shared/output/ -i /Shared/4K/ -b swdl_blocks.json -
s ~/workspace/thick/package/conti/conti-swl/scripts -e
~/workspace/thick/up_environment
```

For the execution as an imported module, the module must me imported:

```
from generate_swdl_blocks import create_swdl_blocks
```

This is the API definition:

```
create_swdl_blocks(output_dir, image_dir, blocks_file, swl_scripts, env)
```

!Note: The parameters must be provided in this specific order

Here is an example of how to invoke it with no optional parameters:

```
#!/usr/bin/env python3

from generate_swdl_blocks import create_swdl_blocks


ret = create_swdl_blocks("/Shared/output/", "/Shared/4K/")
if (ret !=0):
    print("SWDL blocks generation failed")
else:
    print("SWDL Blocks were generated successfully")
```

And this is how to invoke it with optional parameters:

```
#!/usr/bin/env python3

from generate_swdl_blocks import create_swdl_blocks


ret = create_swdl_blocks("/Shared/output/", "/Shared/4K/", "swdl_blocks.json",
"~/workspace/thick/package/conti/conti-swl/scripts", "~/workspace/thick/up_environment")
if (ret !=0):
    print("SWDL blocks generation failed")
else:
    print("SWDL Blocks were generated successfully")
```


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Once the script execution finishes, the output directory will contain one update package (iso) per block.

Name	Size	Type	Modified
Block_1.iso	550.9 kB	Unknown	19:38
Block_2.iso	11.4 MB	Unknown	19:37
Block_3.iso	128.9 MB	Unknown	19:38

Figure 25 Expected output

These iso files are intended to be used to perform an update using SWL capability. For more information please refer to document TP_F101U-00028.

3.1.3.3 Testing/Error Reporting

Not Applicable

4 Known Issues

Not Applicable

5 FAQs

Not Applicable

6 Troubleshooting/Tips and Tricks

Not Applicable

7 References

7.1 Abbreviations/Acronyms/Definitions

Not Applicable

7.2 Platform Documentation

TP_102G-00001: Telematics Platform Glossary
TP_F101U-00028: Software Loading User Guide
TP_F101U-00035: Software Development Kit User Guide
TP_F100W-00015: Update Agent Writers Guide
TP_F101U-00040: Update Package Generation User Guide

7.3 Website links

https://www.json.org/

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8 Revision History

Version	Date	Description
01.00	Oct 01, 2021	Initial Release

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